Code-B Phase-1

Aakash

Medical|IIT-JEE|Foundations

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MM : 45 | Subjective Test_OYMP1-2526_PST-02B | Time : 90 Min.

Topics Covered:
Physics: Current Electricity, Moving Charges and Magnetism
Chemistry: Electrochemistry, Chemical Kinetics
Biology: Principles of Inheritance & Variation, Reproductive Health

PHYSICS

Q1. If the length of a conductor is tripled, then its resistivity will [1]

(1) Be doubled
(2) Be halved
(3) Be tripled
(4) Remain same

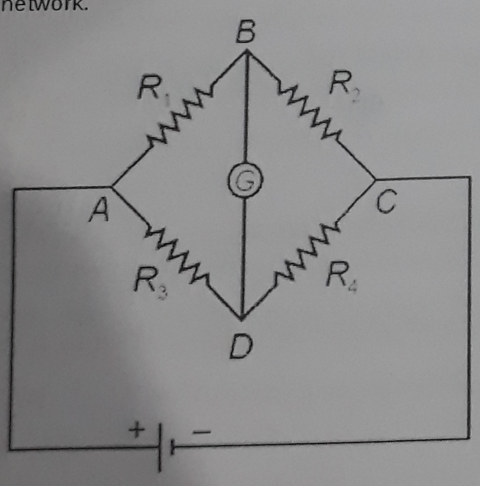
Q2. On increasing the temperature resistance of conductors ___. [1]

Q3. Does a magnetic field exert a force on a static charge? Explain. [1]

Q4. An electron and proton enter perpendicular to a uniform magnetic field with the same speed. How many times larger will the radius of proton's path be than the electron's path? Proton is 1840 times heavier than electron. [2]

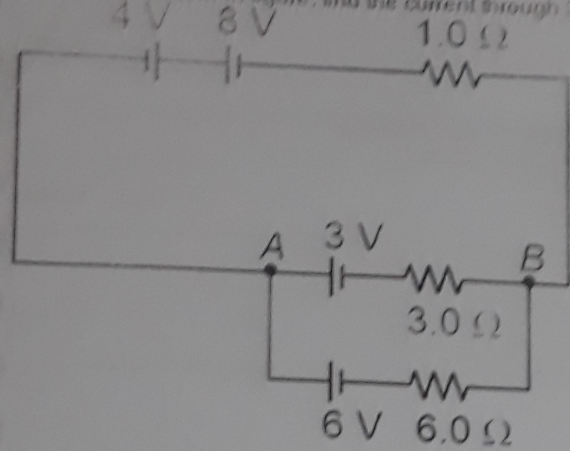
Q5. Derive a relation between drift velocity of free electrons and electric current. [2]

Q6. The given figure shows a network of resistances R_1, R_2, R_3 and R_4. Using Kirchhoff's laws, establish the balance condition for the network. [3]



OR

In the circuit shown in figure, find the current through 3 Ω and 6 Ω resistor. [3]



Q7. Derive an expression for the magnetic field along the axis of an air cored solenoid, using Ampere's circuital law. Sketch the magnetic field lines for a finite solenoid. Explain why the field at the exterior mid point is weak while at the interior it is uniform and strong. [5]

OR

State the Biot Savart's Law for the magnetic field due to a current carrying element. Use this law to obtain a formula for magnetic field at the centre of circular loop of radius R carrying a steady current I. Sketch the magnetic field lines for the current loop clearly indicating the direction of the field. [5]

CHEMISTRY

Q8. Explain, why aqueous solution of $CuSO_4$ turns colourless on complete electrolysis using platinum electrodes. [1]

Q9. The slope of Arrhenius plot $\left(\ln k \text{ vs } \frac{1}{T}\right)$ of first order reaction is -2×10^3 K. The value of E_a of the reaction is [1]

(1) -83 kJ mol^{-1} (2) 16.6 kJ mol^{-1}

(3) 83.0 kJ mol^{-1} (4) 166 kJ mol^{-1}

Q10. The standard electrode potential for Daniell cell is 1.1 V. Calculate the standard Gibbs energy for the cell reaction. [2]

$Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$ (F = 96500 C mol^{-1})

Q11. A first order decomposition reaction takes 40 minutes for 30% decomposition. Calculate $t_{1/2}$ value. [Given $\log_{10}\left(\frac{10}{7}\right) = 0.155$] [2]

Q12(i) In a reaction between A and B, the initial rate (r_0) was measured at different initial concentration of A and B given below. [3]

A $(mol\ L^{-1})$	B $(mol\ L^{-1})$	r_0 $(mol\ L^{-1} s^{-1})$
0.1	0.1	2×10^{-4}
0.2	0.1	8×10^{-4}
0.2	0.2	8×10^{-4}

Calculate order with respect to A, B and rate constant of the reaction.

(ii) For the reaction $A + 2B \rightarrow C$, the rate of disappearance of B is 0.2×10^{-4} mol L^{-1} s^{-1}. Find the rate of reaction. [1]

OR

Define Kohlrausch law. Λ_m° for NaCl, HCl and CH_3COONa respectively are 126, 426 and 91 S cm^2 mol^{-1}. Calculate Λ° for CH_3COOH. [4]

Q13(i) A reaction is third order with respect to a reactant. How is the rate of reaction affected if concentration of the reactant is [3]
(i) Doubled
(ii) Reduced to one half

(ii) Given that, the standard electrode potential (E°) of metals are [2]
$Li^+/Li = -3.05$ V
$Al^{3+}/Al = -1.66$ V
$Fe^{2+}/Fe = -0.44$ V
$Sn^{2+}/Sn = -0.14$ V
Arrange these metals in an increasing order of their reducing power.

OR

(i) Calculate the emf of the cell in which the following reaction takes place: [3]
$Ni(s) + 2Ag^+(0.002\ M) \rightarrow Ni^{2+}(0.160\ M) + 2Ag(s)$
Given that $E^\circ_{cell} = 1.05$ V, $\log_{10}2 = 0.3$

(ii) Write two differences between 'order of reaction' and 'molecularity of reaction'. [2]

BIOLOGY

Q14. Experimental verification of the chromosomal theory of inheritance was done by [1]

(1) Walter Sutton
(2) Theodore Boveri
(3) T.H. Morgan
(4) Alfred Sturtevant

OR

A method of sterilization in which a part of fallopian tube is cut or tied up is [1]

(1) Called vasectomy
(2) A barrier method of contraception
(3) A method of inhibition of ovulation
(4) Highly effective for birth control

Q15. Define pleiotropic gene with the help of example. [2]

Q16. (a) Name the institute where 'Saheli', a non-steroidal oral contraceptive pill, was developed. [2]
(b) Write one disadvantage and one advantage of amniocentesis.

Q17. What is Klinefelter's syndrome? Write down the features of the person inflicted with Klinefelter's syndrome? [2]

OR

(a) Write down four features of an ideal contraceptive. [2]
(b) Mention two examples each of copper releasing and hormone releasing IUDs.

Q18. A man with normal eye sight and no history of colourblindness in his family marries a normal woman whose father was colourblind. Work out the cross and the probability of their son being colourblind. [3]

OR

(i) Define sexually transmitted infections. Name sexually transmitted infections and which of them are not curable? [2]

(ii) How does lactational amenorrhea account for birth control? [1]

Q19. (a) Write down the genotype of a plant which can produce eight different types of gametes. [5]
(b) Which blood groups are possible in progeny if parents have blood group AB and O?
(c) How many kind of phenotypes would you expect in F_2 generation in a monohybrid cross if flower colour of snapdragon is considered?
(d) Briefly explain the pattern of inheritance, where F_1 phenotype resembles only one of the two parents. Also, name the patten of inheritance.
(e) Explain male heterogamety.

OR

(i) Expand: AI and ICSI [1]

(ii) How do 'implants' act as an effective method of contraception in human females? Mention its one advantage over contraceptive pills. [2]

(iii) Enumerate and describe any four reasons for introducing sex education to school-going children. [2]